Ratio & Proportion

Year 4 · **Mark Scheme**

Lesson Worksheet — Answers

Easy / Standard

1. 7 × 2 = 14

2. C — 6:8

3. 4 × 3 = 12

4. A, C and E

5. 2 × 4 = 8

6. 8 × 4 = 32

7. B — 8:16

8. 3 × 4 = 12

9. A, C and E

10. 3 × 3 = 9

11. 2 × 4 = 8

12. C — 16:12

13. 3 × 5 = 15

14. 2 each

15. 3 × 4 = 12

Standard

16. Parts = 2; each part = 2; apples = 1 × 2 = 2

17. C — 10g

18. Per person = 6g; for 6 people = 36g

19. A, B and D

20. 4 × 17 = 68 cm

21. Parts = 3; each part = 5; apples = 1 × 5 = 5

22. C — 40g

23. Per person = 3g; for 9 people = 27g

24. 12 and 12

25. 4 × 18 = 72 cm

26. Parts = 3; each part = 4; apples = 2 × 4 = 8

27. B — 15g

28. A, C and E

29. 2 × 20 = 40 cm

30. Per person = 2.5g; for 6 people = 15g

Standard / Hard

31. C — 24g (8g ÷ 2 = 4g per person; 4g × 6 = 24g)

32. Price per pencil = £3; 5 pencils = £15

33. Rate = 3 litres per minute; 3 × 5 = 15 litres

34. 8 parts total; each part = 5; larger share = 5 × 5 = 25

35. 4 parts total; each part = 5ml. Juice = 5ml; water = 15ml

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36. C — 25g (10g ÷ 2 = 5g per person; 5g × 5 = 25g)

37. Price per pencil = £5; 6 pencils = £30

38. A, B and D (rate = 9 ÷ 3 = 3 litres per minute; 3 × 5 = 15 litres)

39. 5 parts total; each part = 6; larger share = 6 × 3 = 18

40. 3 parts total; each part = 3ml. Juice = 3ml; water = 6ml

41. B — 28g (8g ÷ 2 = 4g per person; 4g × 7 = 28g)

42. Price per pencil = £3; 8 pencils = £24

43. Rate = 4 litres per minute; 4 × 6 = 24 litres

44. 5 parts total; each part = 4; larger share = 4 × 4 = 16

45. A, B, C and E (5 parts; each part = 4ml; juice = 2 × 4 = 8ml; water = 3 × 4 = 12ml; D is wrong since juice is 8ml, not 10ml)

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Homework Worksheet — Answers

Easy / Standard

1. B — 6:4

2. A, C and E

3. 8 × 4 = 32

4. 3 × 5 = 15

5. 3 × 4 = 12

6. C — 16:16

7. 3 each

8. 6 × 3 = 18

9. 3 × 3 = 9

10. 3 × 6 = 18

11. B — 12:6

12. A, C and E

13. 7 × 2 = 14

14. 2 × 3 = 6

15. 1 × 5 = 5

Standard

16. A, C and E

17. Parts = 2; each part = 2; apples = 1 × 2 = 2

18. C — 80cm

19. Per person = 3g; for 9 people = 27g

20. Per person = 3g; for 3 people = 9g

21. 9 and 9

22. Parts = 2; each part = 3; apples = 1 × 3 = 3

23. C — 85cm

24. Per person = 3g; for 6 people = 18g

25. 20g

26. A, C and E

27. Parts = 4; each part = 4; apples = 3 × 4 = 12

28. B — 36cm

29. 36g

30. 30g

Standard / Advanced

31. A — 0 (the ratio 1:1 means the groups are equal)

32. 4 × 5 = 20 cakes

33. B — 2:3

34. 4 parts total; each part = 5. Shares: 15 and 5; difference = 10

35. Scale factor = 2; flour = 6 × 2 = 12g

36. A, B, C and E (total time = 30 ÷ 5 = 6 minutes; time passed = 2 minutes; more minutes needed = 4; more litres needed = 30 − 10 = 20)

37. 2 parts total; each part = 5 litres; red = 5 litres

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38. A and B (scale is 1cm to 20cm; real bridge = 4 × 20 = 80cm)

39. 5 parts each side; apples = 5; cost = 5 × £1 = £5

40. Scale factor = 2; flour = 6 × 2 = 12g

41. Total time needed = 7 minutes; time already passed = 4 minutes; more minutes needed = 3

42. B — 3 litres (3 parts total; each part = 3 litres; red = 1 × 3 = 3 litres)

43. Scale: 1cm model = 15cm real; real bridge = 4 × 15 = 60cm

44. 3 parts total; each part = 3; apples = 3; cost = 3 × £2 = £6

45. 6g × 3 = 18g

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